



**NILKAMAL SCHOOL OF MATHEMATICS,
APPLIED STATISTICS & ANALYTICS**

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Course: MSc. Statistics and Data Science

Subject: Regression Analysis

Assignment 1: Multiple Linear Regression with
Indicator (Dummy) Variables

Assignment 2: Autocorrelation in Time-Series
Regression

Assignment 1: Multiple Linear Regression with Indicator (Dummy) Variables

LINEAR REGRESSION IN PRESENCE OF INDICATOR VARIABLES

Dataset: Hotel Booking Demand | Source: Kaggle

1. Introduction

In regression analysis, we often come across instances when we need to deal with qualitative variables, i.e., variables that are not quantitative in nature. Variables like hotel type, meal plan, or market segment are all qualitative in nature. They are very important but cannot be fed into a regression equation as raw numbers. To feed qualitative variables into a regression equation, we need to create indicator variables, also known as dummy variables. Indicator variables are artificially created variables, usually of binary type, i.e., variables that take 1 or 0.

It is necessary to understand the reasons for which it would be wrong to assign arbitrary numbers to categories, for example, 1 = City Hotel, 2 = Resort Hotel. Indicator variables help us avoid this issue.

Why $(p - 1)$ Dummy Variables?

When a categorical variable has p distinct levels, only $(p - 1)$ indicator variables are included in the model. The excluded level becomes the reference (baseline) category. All dummy coefficients are then interpreted as differences relative to this reference. For a two-level variable like hotel type (City Hotel and Resort Hotel), we define exactly one dummy: $\text{hotel_dummy} = 1$ if Resort Hotel, 0 if City Hotel. City Hotel therefore becomes the reference.

The Dummy Variable Trap

If all p dummies are included, their columns would sum up to 1 for every observation, hence creating a perfect linear relationship with the intercept column. This would result in perfect multicollinearity, and the $(X'X)$ matrix would be singular, hence the OLS estimator would be undefined, a problem that is known as the dummy variable trap. The solution to this problem is to include exactly $(p - 1)$ dummies, allowing the omitted group to be absorbed by the intercept.

- **Handling a Binary Categorical Predictor**

Suppose we are in a situation where the observations are classified as members of exactly two groups, for example, individuals in Group A and individuals in Group B, with a continuous measurement X_1 . We cannot directly enter this category in the regression equation; however, we can represent the category with a single binary variable:

$X_2 = 0 \rightarrow$ observation is in Group A (the reference)

$X_2 = 1 \rightarrow$ observation is in Group B

Group A is designated the reference because its contribution is folded into the intercept of the model. The estimated equation then takes the form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

Substituting each possible value of X_2 separates this single equation into two group-specific lines:

Group A ($X_2 = 0$): $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1$

Group B ($X_2 = 1$): $\hat{Y} = (\hat{\beta}_0 + \hat{\beta}_2) + \hat{\beta}_1 X_1$

- **Extending to Three or More Categories**

Two things to observe. First, observe that the slope for both lines is the same, i.e., $\hat{\beta}_1$. This implies that the rate of change of Y relative to X_1 is assumed to be the same for all groups. Second, observe that the two lines are parallel. This implies that the distance between the two lines is assumed to be constant, i.e., $\hat{\beta}_2$, for any given value of X_1 . If $\hat{\beta}_2$ is positive, Group B is positioned higher, and if $\hat{\beta}_2$ is negative, Group B is positioned lower. This model is called parallel lines/equal slopes.

The logic of the binary dummy variable extends naturally to the case where the categorical variable has more than two categories. The basic rule to remember is: if the categorical variable has p categories, then you should have $p - 1$ dummy variables, and leave out one of the groups. To see this, note that if you include all p dummy variables, the matrix would be rank deficient (the dummy variable trap), so one group has to be omitted.

For a three-group variable (A, B, C), the coding scheme is as follows:

Category	X_2	X_3	What gets switched on
A — Reference	0	0	Neither dummy — Group A is the baseline throughout
B	1	0	X_2 activates, signalling Group B membership
C	0	1	X_3 activates, signalling Group C membership

The regression model is extended to accommodate both dummies

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

Plugging in the dummy values group by group reveals three parallel fitted lines:

$$\text{For Group A } (X_2=0, X_3=0): \hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1$$

$$\text{For Group B } (X_2=1, X_3=0): \hat{Y} = (\hat{\beta}_0 + \hat{\beta}_2) + \hat{\beta}_1 X_1$$

$$\text{For Group C } (X_2=0, X_3=1): \hat{Y} = (\hat{\beta}_0 + \hat{\beta}_3) + \hat{\beta}_1 X_1$$

The slope $\hat{\beta}_1$ is common for all three lines, only the intercept is different. $\hat{\beta}_2$ represents the difference between Group B's mean Y and Group A's mean Y, given that X_1 is held constant, and $\hat{\beta}_3$ represents the same for Group C. For any given p-level categorical variable, the same principle is applied, where each additional dummy has the same slope, but a different shift in the intercept based on its own difference.

2. Dataset Overview

The Hotel Booking Demand dataset is available on Kaggle and provides booking information for two different types of hotels in Portugal: a city hotel and a resort hotel, for check-ins between 2015 and 2017. The data consists of one row for each booking. This dataset is often used for research in hotel analytics.

Dataset Link:- <https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>

Attribute	Details
Total Observations	119,390 booking records
Total Variables	32 columns (mix of numerical and categorical)
Dependent Variable (Y)	adr — Average Daily Rate (USD per night)
Hotel Types	City Hotel (reference) and Resort Hotel
Time Period	2015, 2016, and 2017
Key Numerical Predictors	lead_time, stays_in_week_nights, previous_bookings_not_canceled
Key Categorical Predictors	hotel (converted to dummy: hotel_dummy)
Missing Values	4 missing values in children column (imputed with mean)

The dependent variable 'adr', or Average Daily Rate, represents the average revenue generated by each occupied room on a daily basis. The key modeling question of interest here is whether the type of hotel, i.e., Resort or City, impacts the rate in a significant manner, while controlling for the booking lead time and length of stay.

3. Data Preprocessing

Step 1: Load Library and Data

```
# LOAD LIBRARY
library(readr)

## Warning: package 'readr' was built under R version 4.4.1

# LOAD DATA
df <- read_csv("C:/Users/omkar
navle/Desktop/hotel_bookings.csv/hotel_bookings.csv")

## Rows: 119390 Columns: 32
## — Column specification
## Delimiter: ","
## chr (13): hotel, arrival_date_month, meal, country, market_segment,
distrib...
## dbl (18): is_canceled, lead_time, arrival_date_year,
arrival_date_week_numb...
## date (1): reservation_status_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this
message.

View(df)
str(df)

## spc_tbl_ [119,390 × 32] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ hotel : chr [1:119390] "Resort Hotel" "Resort
Hotel" "Resort Hotel" "Resort Hotel" ...
```

Step 2: Summary Statistics

SUMMARY STATISTICS

```
summary(df)

##      hotel      is_canceled      lead_time      arrival_date_year
## Length:119390   Min.   :0.0000   Min.    :  0   Min.    :2015
## Class :character 1st Qu.:0.0000   1st Qu.: 18   1st Qu.:2016
## Mode  :character Median :0.0000   Median : 69   Median :2016
##              Mean  :0.3704   Mean  :104   Mean  :2016
##              3rd Qu.:1.0000   3rd Qu.:160   3rd Qu.:2017
##              Max.   :1.0000   Max.   :737   Max.   :2017
##
## arrival_date_month arrival_date_week_number arrival_date_day_of_month
## Length:119390      Min.    : 1.00      Min.    : 1.0
## Class :character   1st Qu.:16.00      1st Qu.: 8.0
## Mode  :character   Median :28.00      Median :16.0
##              Mean  :27.17      Mean  :15.8
##              3rd Qu.:38.00      3rd Qu.:23.0
##              Max.   :53.00      Max.   :31.0
##
## stays_in_weekend_nights stays_in_week_nights      adults
## Min.   : 0.0000      Min.    : 0.0      Min.    : 0.000
## 1st Qu.: 0.0000      1st Qu.: 1.0      1st Qu.: 2.000
## Median : 1.0000      Median : 2.0      Median : 2.000
## Mean    : 0.9276      Mean    : 2.5      Mean    : 1.856
## 3rd Qu.: 2.0000      3rd Qu.: 3.0      3rd Qu.: 2.000
## Max.    :19.0000      Max.    :50.0      Max.    :55.000
##
##
```

Step 3: Check and Handle Missing Values

CHECK MISSING VALUES

```
sapply(df, function(x) sum(is.na(x)))

##      hotel      is_canceled
##      0          0
##      lead_time      arrival_date_year
##      0          0
##      arrival_date_month      arrival_date_week_number
##      0          0
##      arrival_date_day_of_month      stays_in_weekend_nights
##      0          0
##      stays_in_week_nights      adults
##      0          0
##      children      babies
##      4          0
##      meal      country
##      0          0
##      market_segment      distribution_channel
##      0          0
##      is_repeated_guest      previous_cancellations
##      0          0
##      previous_bookings_not_canceled      reserved_room_type
##      0          0
##      assigned_room_type      booking_changes
##      0          0
##      deposit_type      agent
##      0          0
##      company      days_in_waiting_list
##      0          0
##      customer_type      adr
##      0          0
##      required_car_parking_spaces      total_of_special_requests
##      0          0
##      reservation_status      reservation_status_date
##      0          0
```

```

# HANDLE MISSING VALUES
df$children[is.na(df$children)] <- mean(df$children, na.rm = TRUE)
df$country[is.na(df$country)] <- "Unknown"
df$agent[is.na(df$agent)] <- 0
df$company[is.na(df$company)] <- 0

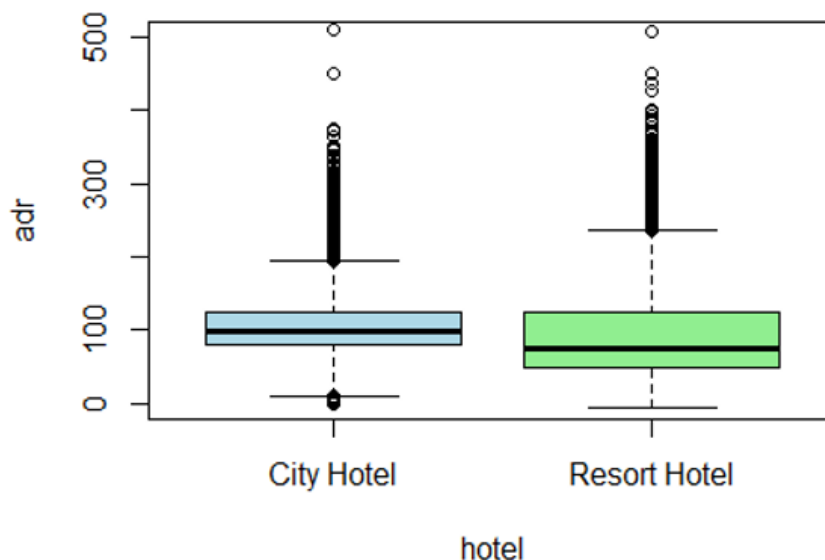
sapply(df, function(x) sum(is.na(x)))

##          hotel          is_canceled
##          0          0
##      lead_time    arrival_date_year |
##          0          0
##    arrival_date_month    arrival_date_week_number
##          0          0
##    arrival_date_day_of_month    stays_in_weekend_nights
##          0          0
##      stays_in_week_nights          adults
##          0          0
##          children          babies
##          0          0
##          meal          country
##          0          0
##      market_segment    distribution_channel
##          0          0
##      is_repeated_guest    previous_cancellations
##          0          0
##    previous_bookings_not_canceled    reserved_room_type
##          0          0
##      assigned_room_type    booking_changes
##          0          0
##      deposit_type          agent
##          0          0
##          company    days_in_waiting_list
##          0          0
##      customer_type          adr
##          0          0
##    required_car_parking_spaces    total_of_special_requests
##          0          0
##      reservation_status    reservation_status_date
##          0          0

```


The qualitative variable hotel has two categories, namely 'City Hotel' and 'Resort Hotel'. Hence, as $p = 2$, we need to define exactly $p - 1 = 1$ dummy variable. City Hotel is taken as the reference category and its effect is included in the intercept of the model. The coefficient of the dummy tells us how the ADR of a Resort Hotel differs from a City Hotel, controlling for all other variables.

ADR by Hotel Type



5. Full OLS Model

The full model regresses `adr` on `lead_time`, `stays_in_week_nights`, `previous_bookings_not_canceled`, and the hotel dummy variable. This gives a comprehensive initial picture of which predictors matter.

```
# FULL MODEL
model <- lm(adr ~ lead_time + stays_in_week_nights
+previous_bookings_not_canceled + hotel_dummy,data = df)

summary(model)

##
## Call:
## lm(formula = adr ~ lead_time + stays_in_week_nights +
##     previous_bookings_not_canceled +
##     hotel_dummy, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -206.7   -31.1    -8.2    22.0  5294.3
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    104.388409    0.273429   381.77 <2e-16 ***
## lead_time       -0.045422    0.001379   -32.94 <2e-16 ***
## stays_in_week_nights    2.853154    0.079093    36.07 <2e-16 ***
## previous_bookings_not_canceled -2.476605    0.096486   -25.67 <2e-16 ***
## hotel_dummy     -13.789998    0.315908   -43.65 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.75 on 119385 degrees of freedom
## Multiple R-squared:  0.03084,    Adjusted R-squared:  0.03081
## F-statistic: 949.9 on 4 and 119385 DF,  p-value: < 2.2e-16
```

All predictors of the full model are statistically significant at a 1% significance level ($p < 2e-16$). The F-statistic of 949.9 also proves the joint significance of the regressors. The R-squared of 0.031 suggests moderate explanatory power of the model, which is a common feature of booking data sets due to their variability.

6. Model Refinement

After reviewing the full model, the feature `previous_bookings_not_canceled` is removed to yield a more parsimonious model. Even though it was statistically significant, this feature has a relatively minor practical contribution to the model compared to the cleaner three-predictor model. The improved model includes the three most interpretable and practically significant predictors.

```
# REFINED MODEL
model_refined <- lm(adr ~ lead_time + stays_in_week_nights + hotel_dummy, data =
df)

summary(model_refined)

##
## Call:
## lm(formula = adr ~ lead_time + stays_in_week_nights + hotel_dummy,
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -209.4   -31.3    -8.3    22.2  5294.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    103.63928     0.27261   380.17  <2e-16 ***
## lead_time       -0.04312     0.00138   -31.25  <2e-16 ***
## stays_in_week_nights  2.93050     0.07925    36.98  <2e-16 ***
## hotel_dummy     -13.85879     0.31677   -43.75  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.89 on 119386 degrees of freedom
## Multiple R-squared:  0.02549,    Adjusted R-squared:  0.02547
## F-statistic: 1041 on 3 and 119386 DF,  p-value: < 2.2e-16
```

7. Interpretation of Coefficients

The refined model gives rise to two separate regression equations one for each hotel type that share a common slope but differ in their intercepts. This is the hallmark of a dummy variable model.

For City Hotel ($\text{hotel_dummy} = 0$): $\text{ADR} = 103.64 - 0.043 \times \text{lead_time} + 2.931 \times \text{stays_in_week_nights}$

For Resort Hotel ($\text{hotel_dummy} = 1$): $\text{ADR} = (103.64 - 13.86) - 0.043 \times \text{lead_time} + 2.931 \times \text{stays_in_week_nights}$

$$= 89.78 - 0.043 \times \text{lead_time} + 2.931 \times \text{stays_in_week_nights}$$

Variable	Coefficient	Interpretation
(Intercept)	103.639	Baseline ADR for a City Hotel booking with zero lead time and no weeknight stays
lead_time	-0.043	Each additional day of advance booking reduces ADR by about \$0.04 — consistent with early-bird discounting
stays_in_week_nights	2.931	Each additional weeknight stay adds approximately \$2.93 to the average daily rate
hotel_dummy (Resort Hotel)	-13.859	Resort Hotels charge \$13.86 LESS per night than City Hotels on average, after controlling for lead time and length of stay

The negative coefficient for hotel_dummy is initially puzzling, as resort hotels are generally considered high-end properties. However, once controlling for chemical lead time and length of stay, the data suggests that city hotels in this sample command a higher average daily rate. This may be due to the urban business travel market, in which city hotels cater to corporate customers willing to pay a premium for a shorter, last-minute stay, as indicated by the positive coefficient for $\text{stays_in_week_nights}$ and the negative coefficient for lead_time .

The negative sign of the lead_time effect simply verifies the well-known early bird discount principle, whereby customers who book early are rewarded with a cheaper price. This is consistent with the dynamic pricing strategy, which has been widely adopted in the hotel and airline industries.

Assignment 2: Autocorrelation in Time-Series Regression

Dataset: Walmart Sales Dataset | Source: Kaggle

1. Introduction to Autocorrelation

- **What is Autocorrelation?**

The Classical Linear Regression Model (CLRM) assumes that error terms between different observations are uncorrelated. Mathematically:

- $\text{Cov}(u_i, u_r) = 0$ for every pair $i \neq j$

Autocorrelation, also known as Serial Correlation, is a violation of this assumption. It is present when there is a systematic link between errors of two different observations. This means that in most cases, errors of consecutive time periods have a common sign: if an error is positive in period t , then an error in period $t+1$ will be positive. **Root Causes**

- **Economic Inertia and Momentum**

The major macroeconomic variables such as national income, employment, industrial production, and money supply do not exhibit erratic movements. Rather, they show trends over a period of time. An economy emerging from a recession will show a growth rate over a series of consecutive quarters. Conversely, a recession will show a decline over a series of consecutive quarters. This multi-period nature of the macroeconomic variables ensures that each error term includes a carryover from the last period.

- **Omitted Variable Bias**

Each time a relevant variable is omitted, its effect enters the error term. The problem is that if the omitted variable has its own trend or cycle, as most economic series do, the residuals will have a pattern rather than true random noise.

- **Supply-Demand Lags: The Cobweb Effect**

In markets with production lead times, particularly agriculture, suppliers must commit to output quantities before knowing the final price. They base decisions on last season's price, leading to the supply specification $S_t = \beta_1 + \beta_2 P_{t-1} + u_t$. A glut in year t depresses prices, inducing under-supply in year $t+1$, which then pushes prices up again. This alternating over- and under-production imprints a cyclical cobweb pattern directly onto the disturbance terms.

- **Temporal Aggregation and Smoothing**

When high-frequency raw data, for example, daily or monthly, are aggregated or averaged to form lower frequency series, for example, quarterly or yearly, the aggregated or averaged adjacent observations overlap, as they are made up of overlapping high-frequency raw observations.

This artificial smoothness occurs because, even with random high-frequency raw data, aggregated or averaged adjacent observations are not independent of each other.

Though autocorrelation is mostly a time-series problem, it can also arise in cross-sections. If neighbouring spatial units, like households on the same street or firms in the same industry cluster, react similarly to local shocks, then spatial correlation is equivalent to time-series autocorrelation..

- **What Goes Wrong When Autocorrelation is Ignored**

Autocorrelation does not bias the OLS coefficient estimates ,they remain unbiased and consistent. The damage is subtler and concentrated in the standard errors and the efficiency of estimation:

Problem	Mechanism	Practical Consequence
OLS loses efficiency	Estimators are no longer minimum-variance among linear unbiased estimators	BLUE status is lost; GLS outperforms OLS
Standard errors shrink artificially	OLS variance formula omits the autocorrelation terms, producing downward bias	Coefficients look more precise than they truly are
t and F tests break down	Inflated t-statistics caused by underestimated standard errors	Spurious significance: predictors falsely appear to matter
R^2 is overstated	RSS/df underestimates the true variance of disturbances	Model fit appears better than it genuinely is

- **Detecting Autocorrelation**

Visual Inspection

A plot of OLS residuals against the time index, also called a time sequence plot, should be the first approach. In a situation with real independence, the residuals should be randomly scattered above and below zero with no apparent trend or wave pattern. Positive autocorrelation should be evident as long stretches, where the series stays above zero for a long time, then stays below zero for a long time as well. Conversely, negative autocorrelation should be evident as rapid sign changes, or a square wave pattern.

A similar test is the lag residual scatter plot. This involves a graphical method of plotting e_t against e_{t-1} . The presence of a positive linear trend indicates positive autocorrelation. A negative trend indicates negative autocorrelation. A shapeless distribution indicates independence. This method is quick and subjective. It is therefore always important to perform a formal test.

- **Durbin-Watson Test**

The Durbin-Watson (DW) test provides a formal, widely-used check for first-order serial correlation. Its statistic is built from consecutive residual differences:

$$d = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2}$$

Because $d \approx 2(1 - \rho)$, where ρ is the first-order autocorrelation coefficient, the statistic carries intuitive meaning across its 0–4 range:

d Range	Signal	Diagnosis
Near 0	ρ close to +1	Strong positive autocorrelation
Near 2	ρ close to 0	Residuals appear uncorrelated — CLRM assumption likely holds
Near 4	ρ close to –1	Strong negative autocorrelation
$d_l < d < d_u$	Indeterminate	Test is inconclusive; seek additional evidence

The DW test is valid only when: (1) an intercept is present in the model, (2) regressors are non-stochastic, (3) errors follow AR(1) structure, and (4) no lagged dependent variable appears as a regressor. Violating any of these conditions makes the statistic unreliable.

- **Breusch-Godfrey (BG) / LM Test**

The Breusch-Godfrey test removes the main restrictions of the DW test. It handles higher-order AR schemes, lagged dependent variable regressors, and moving-average error structures. The four-step procedure is:

Step 1. Run OLS on the original model and save the residuals e_t .

Step 2. Auxiliary regression: regress e_t on its p lags ($e_{t-1}, e_{t-2}, \dots, e_{t-p}$) together with all original regressors.

Step 3. Form the test statistic $(n - p) \times R^2$ from the auxiliary regression, which is asymptotically χ^2 distributed with p degrees of freedom.

Step 4. Reject H_0 (no autocorrelation) if the statistic exceeds χ^2_p at the chosen significance level.

- **Correcting for Autocorrelation**

- **Cochrane-Orcutt Transformation When ρ is Known**

The most principled remedy transforms the original model to produce classical (non-autocorrelated) errors. Starting from the two-variable model $Y_t = \beta_1 + \beta_2 X_t + u_t$ and its one-period lag, multiply the lagged equation by ρ and subtract:

Quasi-differenced outcome: $Y_t^* = Y_t - \rho Y_{t-1}$

Quasi-differenced predictor: $X_t^* = X_t - \rho X_{t-1}$

Transformed intercept: $\beta_1^* = \beta_1(1 - \rho)$

Transformed error: $v_t = u_t - \rho u_{t-1} \rightarrow \text{white noise}$

OLS applied to the transformed variables Y_t^* and X_t^* delivers GLS estimators that are BLUE. This quasi-differencing approach is the Cochrane-Orcutt transformation.

- **Estimating ρ When it is Unknown**

Method 1 From the DW statistic: Since $d \approx 2(1 - \rho)$, a quick plug-in estimate is $\hat{\rho} = 1 - d/2$. Convenient but only reliable in large samples.

Method 2 —Auxiliary residual regression: Regress e_t on e_{t-1} (no intercept): $e_t = \hat{\rho} e_{t-1} + v_t$. The slope gives a consistent, if slightly small-sample-biased, estimate.

- **Iterative Cochrane-Orcutt , When ρ Needs Refinement**

When the initial $\hat{\rho}$ is unreliable, the estimate is refined through iteration until it converges:

Fit the original model by OLS; extract residuals e_t .

Regress e_t on e_{t-1} to obtain first $\hat{\rho}$.

Construct quasi-differenced series Y_t^* and X_t^* using $\hat{\rho}$.

Re-estimate by OLS; extract new residuals.

Update $\hat{\rho}$ from the new residuals.

Repeat until successive $\hat{\rho}$ values differ by less than 0.01 (convergence).

- **First-Difference Transformation , Assuming $\rho = 1$**

A special limiting case arises when ρ is assumed equal to 1 (perfect positive autocorrelation). The Cochrane-Orcutt subtraction then simplifies entirely to working with period-to-period changes:

$$\Delta Y_t = \beta_2 \Delta X_t + v_t$$

where $\Delta Y_t = Y_t - Y_{t-1}$ and similarly for ΔX_t . The intercept vanishes, and the model is estimated purely in first differences. This is also a standard pre-processing step to eliminate stochastic trends.

- **Durbin's h-Statistic , When DW is Invalid**

Once a lagged dependent variable Y_{t-1} enters the regressor set, the DW statistic is biased toward 2 and loses its diagnostic power. Durbin's h-statistic corrects for this bias:

$$h \approx \hat{\rho} \times \sqrt{[n / (1 - n \cdot V(b_3))]}$$

where n = sample size and $V(b_3)$ = estimated variance of the lag coefficient. For large samples, $h \sim N(0, 1)$, so the 5% decision rule is simply: reject H_0 if $|h| > 1.96$.

Note: If $n \cdot V(b_3) \geq 1$, the square root is undefined and h cannot be computed. In that situation, switch to the Breusch-Godfrey test instead. [2. Dataset Overview](#)

The Walmart Sales Dataset contains weekly sales records for 45 Walmart stores across the United States, covering the period from February 2010 to October 2012. Each row corresponds to a specific store's performance in a given week, yielding 6,435 total observations.

DataSet Link:- <https://www.kaggle.com/datasets/yasserh/walmart-dataset>

Variable	Type	Description
Weekly_Sales (Y)	Numerical	Total weekly revenue per store (USD) — Dependent variable
Temperature (X1)	Numerical	Average temperature at the store's location that week (°F)
Fuel_Price (X2)	Numerical	Average fuel price in the store's region (USD per gallon)
CPI (X3)	Numerical	Consumer Price Index — a measure of regional inflation
Unemployment (X4)	Numerical	Regional unemployment rate (%)
Holiday_Flag	Binary	1 if the week includes a major public holiday, 0 otherwise
Store	Integer	Store identifier (1 to 45)
Date	Date	Week ending date — key for establishing time-series order

It is also apparent that the data has a natural time series pattern, as the data points are collected on a weekly basis for individual stores. Retail sales tend to have high inertia, meaning that if a store has a high sales week, the following week will also be a high sales week, and conversely, if the store has a low sales week, the following week will also be low sales. Therefore, autocorrelation of the residuals in the regression model is highly likely and should be tested for formally.

3. Data Loading and Preprocessing

```
# LOAD LIBRARIES

library(readr)

## Warning: package 'readr' was built under R version 4.4.1

library(lmtest)

## Warning: package 'lmtest' was built under R version 4.4.3

## Loading required package: zoo

## Warning: package 'zoo' was built under R version 4.4.3

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric

library(nlme)

## Warning: package 'nlme' was built under R version 4.4.1

# LOAD DATA

data <- read_csv("C:/Users/omkar navle/Desktop/Walmart.csv")

## Rows: 6435 Columns: 8

## — Column specification —————
## Delimiter: ","
## chr (1): Date
## dbl (7): Store, Weekly_Sales, Holiday_Flag, Temperature, Fuel_Price, CPI,
Un...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this m
essage.

View(data)
str(data)
```

SUMMARY STATISTICS

```
summary(data)
```

```
##      Store      Date      Weekly_Sales      Holiday_Flag
## Min.   : 1      Length:6435      Min.   : 209986      Min.   :0.00000
## 1st Qu.:12      Class :character      1st Qu.: 553350      1st Qu.:0.00000
## Median :23      Mode  :character      Median : 960746      Median :0.00000
## Mean   :23                                     Mean  :1046965      Mean   :0.06993
## 3rd Qu.:34                                     3rd Qu.:1420159      3rd Qu.:0.00000
## Max.   :45      Max.   :3818686      Max.   :1.00000
## Temperature      Fuel_Price      CPI      Unemployment
## Min.   : -2.06      Min.   :2.472      Min.   :126.1      Min.   : 3.879
## 1st Qu.: 47.46      1st Qu.:2.933      1st Qu.:131.7      1st Qu.: 6.891
## Median : 62.67      Median :3.445      Median :182.6      Median : 7.874
## Mean   : 60.66      Mean   :3.359      Mean   :171.6      Mean   : 7.999
## 3rd Qu.: 74.94      3rd Qu.:3.735      3rd Qu.:212.7      3rd Qu.: 8.622
## Max.   :100.14      Max.   :4.468      Max.   :227.2      Max.   :14.313
```

DATE CONVERSION + SORT

```
data$Date <- as.Date(data$Date, format = "%d-%m-%Y")
data <- data[order(data$Date), ]
```

HANDLE MISSING VALUES

```
data <- na.omit(data)
colSums(is.na(data))
```

```
##      Store      Date Weekly_Sales Holiday_Flag Temperature Fuel_Pr
ice
##      0      0      0      0      0
0
##      CPI Unemployment
##      0      0
```

4. OLS Regression Model

In the OLS model, Weekly_Sales is regressed against four macroeconomic variables: Temperature, Fuel_Price, CPI, and Unemployment. The variable 'Holiday_Flag' is not included in the model as the focus should be on macroeconomic variables. The dataset is already ordered in a chronological manner, which is important for future analysis using

time series methods.

```
# OLS MODEL

model <- lm(Weekly_Sales ~ Temperature + Fuel_Price + CPI + Unemployment, data
= data)

summary(model)

##
## Call:
## lm(formula = Weekly_Sales ~ Temperature + Fuel_Price + CPI +
##     Unemployment, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -956788 -478860 -115969  394736 2789686
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1743607.6    79553.1   21.918  < 2e-16 ***
## Temperature    -885.7      396.2   -2.235   0.0254 *
## Fuel_Price    -12248.4    15751.8   -0.778   0.4368
## CPI           -1585.8      195.2   -8.126  5.3e-16 ***
## Unemployment -41215.0     3972.7 -10.375  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 557600 on 6430 degrees of freedom
## Multiple R-squared:  0.02433,    Adjusted R-squared:  0.02372
## F-statistic: 40.09 on 4 and 6430 DF,  p-value: < 2.2e-16
```

The OLS model shows that CPI and Unemployment are highly significant negative predictors of weekly sales, while Fuel_Price is not significant ($p = 0.437$). Temperature has a borderline effect. However, the R-squared of just 0.024 is very low, and because the data is ordered chronologically, the standard errors may be seriously understated due to autocorrelation in the residuals. We must investigate this formally before drawing any conclusions.

5. Detection of Autocorrelation

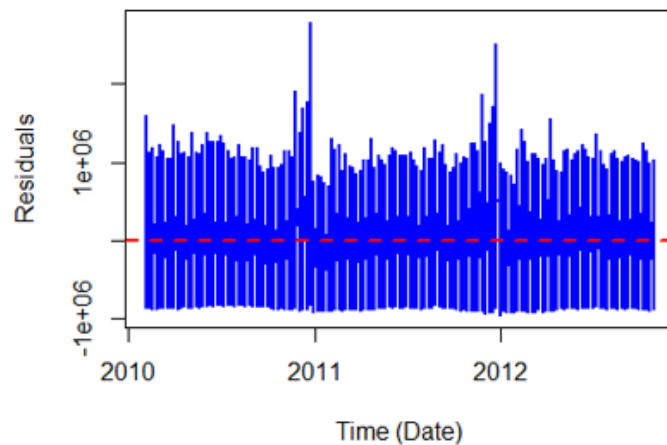
5.1 Graphical Diagnostics

```
# PLOT 1: RESIDUALS OVER TIME
```

```
plot(data$Date, residuals(model), type = "l", col = "blue", lwd = 2, main = "Residuals Over Time", xlab = "Time (Date)", ylab = "Residuals")
```

```
abline(h = 0, col = "red", lwd = 2, lty = 2)
```

Residuals Over Time



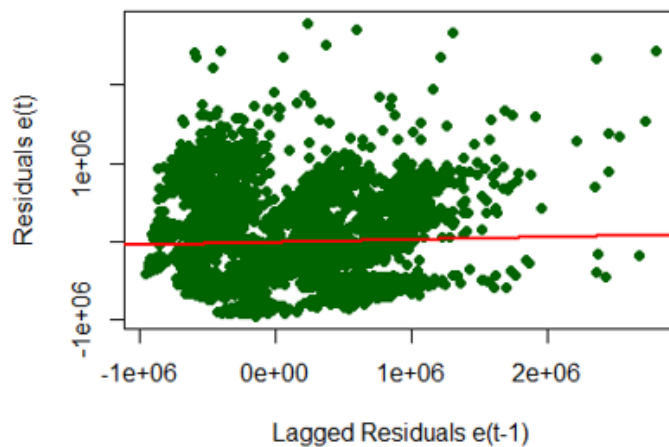
```
# PLOT 2: RESIDUAL LAG PLOT
```

```
e <- residuals(model)
```

```
plot(e[-1], e[-length(e)], col = "darkgreen", pch = 16, main = "Residual Lag Plot", xlab = "Lagged Residuals e(t-1)", ylab = "Residuals e(t)")
```

```
abline(lm(e[-1] ~ e[-length(e)]), col = "red", lwd = 2)
```

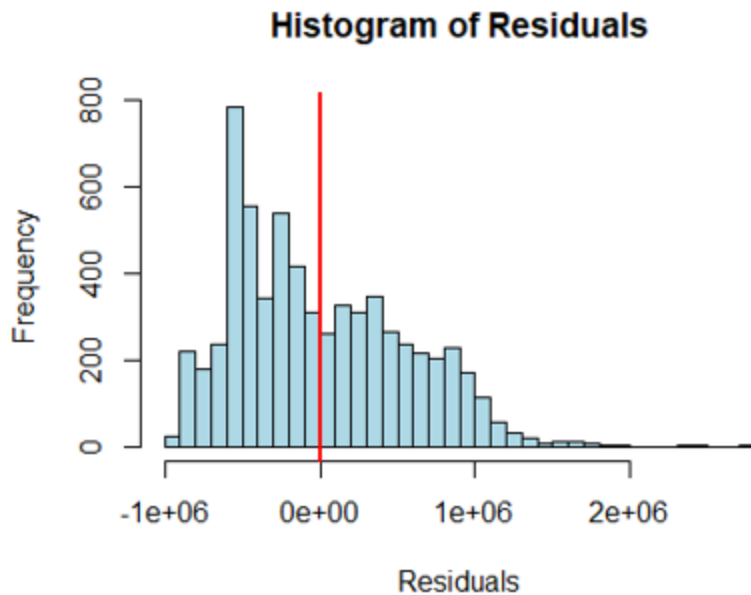
Residual Lag Plot



```
# PLOT 3: HISTOGRAM
```

```
hist(residuals(model),col = "lightblue",border = "black",breaks = 30,main = "Histogram of Residuals",xlab = "Residuals")
```

```
abline(v = 0, col = "red", lwd = 2)
```



From the residual plot over time, we notice that there is a cyclical wave pattern where a cluster of positive residuals occurs consecutively over a period of weeks, then a cluster of negative residuals occurs consecutively. This is the signature plot for a positive autocorrelation effect. From the residual lag plot, we notice that there is a strong positive linear relationship between $e(t)$ and $e(t-1)$.

5.2 Durbin-Watson Test

The DW test is a formal test for the presence of first-order autocorrelation. The null and alternative hypotheses are $H_0: \rho = 0$ (no autocorrelation) versus $H_1: \rho > 0$ (positive autocorrelation). The DW statistic d varies from 0 to 4, and $d \approx 2$ indicates no autocorrelation, $d < 2$ indicates positive autocorrelation, and $d > 2$ indicates negative

autocorrelation.

```
# DURBIN-WATSON TEST

dwtest(model)

##
## Durbin-Watson test
##
## data: model
## DW = 1.9329, p-value = 0.003313
## alternative hypothesis: true autocorrelation is greater than 0
```

Test Component	Value	Interpretation
DW Statistic	1.9329	Below 2 — indicates positive autocorrelation
p-value	0.003313	Highly significant — reject H0 at 1% level
Decision	Reject H0	Statistically significant positive autocorrelation confirmed
Estimated ρ (from residuals)	0.03346	Mild first-order autocorrelation parameter

The Durbin-Watson statistic is 1.9329, which is less than 2. The p-value for the Durbin-Watson test is 0.003313, which is much less than 0.05. We can therefore reject the null hypothesis. Although the DW statistic is only slightly less than 2, indicating mild rather than severe autocorrelation, the result is statistically unambiguous. Any statistical procedure that fails to account for this will not provide reliable standard errors and hypothesis tests.

6. Correcting Autocorrelation: GLS and Cochrane-Orcutt

As autocorrelation is established, we use two different approaches for corrections, which are complementary to each other. One is the Generalized Least Squares model, which can be performed using the nlme package, and the other is the Cochrane Orcutt transformation.

Both of these methods are based on AR(1) errors.

```
# GLS MODEL (AR(1))

gls_model <- gls(Weekly_Sales ~ Temperature + Fuel_Price + CPI + Unemployment
,data = data,correlation = corAR1())

summary(gls_model)

## Generalized least squares fit by REML
## Model: Weekly_Sales ~ Temperature + Fuel_Price + CPI + Unemployment
## Data: data
##      AIC      BIC    logLik
## 188467.1 188514.5 -94226.55
##
## Correlation Structure: AR(1)
## Formula: ~1
## Parameter estimate(s):
##      Phi
## 0.03414656
##
## Coefficients:
##              Value Std.Error   t-value p-value
## (Intercept) 1755066.2  80612.95  21.771517 0.0000
## Temperature   -882.0    406.36  -2.170399 0.0300
## Fuel_Price   -15601.4 16235.29  -0.960956 0.3366
## CPI          -1577.4   194.84  -8.095912 0.0000
## Unemployment -41448.0  3973.74 -10.430483 0.0000
##
## Correlation:
##      (Intr) Tmprtr Fl_Prc CPI
## Temperature  0.007
## Fuel_Price  -0.759 -0.199
## CPI         -0.630 -0.255  0.236
## Unemployment -0.557 -0.184  0.116  0.338
##
## Standardized residuals:
##      Min      Q1      Med      Q3      Max
## -1.7172007 -0.8585049 -0.2086520  0.7068236  5.0016660
##
## Residual standard error: 557642.5
## Degrees of freedom: 6435 total; 6430 residual
```

Manual Cochrane-Orcutt Transformation

```
# ESTIMATE RHO

rho <- sum(e[-1]*e[-length(e)]) / sum(e[-length(e)]^2)
rho

## [1] 0.03346042

# MANUAL COCHRANE-ORCUTT

Y <- data$Weekly_Sales
X1 <- data$Temperature
X2 <- data$Fuel_Price
X3 <- data$CPI
X4 <- data$Unemployment

# Transform variables
Y_star <- Y[-1] - rho * Y[-length(Y)]

X1_star <- X1[-1] - rho * X1[-length(X1)]
X2_star <- X2[-1] - rho * X2[-length(X2)]
X3_star <- X3[-1] - rho * X3[-length(X3)]
X4_star <- X4[-1] - rho * X4[-length(X4)]

# Run transformed regression
model_gls_manual <- lm(Y_star ~ X1_star + X2_star + X3_star + X4_star)

summary(model_gls_manual)

##
## Call:
## lm(formula = Y_star ~ X1_star + X2_star + X3_star + X4_star)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -943659 -477939 -104261  399922 2707913
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1695279.6    77895.2   21.764 < 2e-16 ***
## X1_star      -876.3       406.2   -2.157  0.031 *
## X2_star     -15218.1    16227.0   -0.938  0.348
## X3_star     -1580.2       194.9   -8.110 6.02e-16 ***
## X4_star     -41467.2    3973.7  -10.435 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 557300 on 6429 degrees of freedom
## Multiple R-squared:  0.02439,    Adjusted R-squared:  0.02378
## F-statistic: 40.18 on 4 and 6429 DF,  p-value: < 2.2e-16
```

7. OLS vs. GLS Comparison

```
# OLS vs GLS COMPARISON

ols_coef <- coef(summary(model))
gls_coef <- summary(gls_model)$tTable

comparison <- data.frame(Variable = rownames(ols_coef), OLS_Coeff = ols_coef[,
"Estimate"], GLS_Coeff = gls_coef[, "Value"])

comparison

##           Variable      OLS_Coeff      GLS_Coeff
## (Intercept) (Intercept) 1743607.6200 1755066.1684
## Temperature Temperature   -885.6699   -881.9695
## Fuel_Price   Fuel_Price  -12248.4245 -15601.3934
## CPI          CPI        -1585.8180  -1577.3980
## Unemployment Unemployment -41214.9873 -41448.0283
```

Variable	OLS Coefficient	GLS Coefficient	Key Change
(Intercept)	1,743,607.6	1,755,066.2	Slight increase in baseline
Temperature	-885.7	-882.0	Remains negative and significant
Fuel_Price	-12,248.4	-15,601.4	Magnitude increases; still insignificant
CPI	-1,585.8	-1,577.4	Very stable — robust predictor
Unemployment	-41,215.0	-41,448.0	Highly stable — strongest predictor

As can be seen from the output above, the application of the GLS correction has indeed confirmed the statistical insignificance of Fuel_Price ($p = 0.337$), while CPI and Unemployment continue to be the key statistically significant variables. The Phi value of 0.034 for the AR(1) serial correlation, as estimated by the GLS model, is consistent with the DW statistic of 1.93. The corrected standard errors are slightly larger than the uncorrected standard errors from the OLS model.

8. Refined GLS Model

Fuel_Price is consistently insignificant across both OLS and GLS models. Removing it yields a more parsimonious model that retains all significant predictors.

```
# REFINED MODEL
model_gls_manual_refined <- lm(Y_star ~ X1_star + X2_star + X4_star)

summary(model_gls_manual_refined)

##
## Call:
## lm(formula = Y_star ~ X1_star + X2_star + X4_star)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -885746 -481902 -71738  382431 2736202
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1297302.2    60797.4   21.338 < 2e-16 ***
## X1_star      -1717.0       394.7   -4.350 1.38e-05 ***
## X2_star      15848.0     15847.6    1.000  0.317
## X4_star     -30554.9       3757.7  -8.131 5.06e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 560100 on 6430 degrees of freedom
## Multiple R-squared:  0.01441,    Adjusted R-squared:  0.01395
## F-statistic: 31.34 on 3 and 6430 DF,  p-value: < 2.2e-16
```

9. Summary and Conclusions

Step	Action	Finding
1	OLS Regression	CPI and Unemployment significant; Fuel_Price not significant; DW test needed
2	Residual Plots	Time plot shows cyclical clustering; lag plot shows positive linear pattern
3	Durbin-Watson Test	DW = 1.9329, $p = 0.003$ — statistically significant positive autocorrelation
4	GLS (nlme)	AR(1) $\Phi = 0.034$; corrected standard errors; Fuel_Price remains insignificant
5	Cochrane-Orcutt	Manual transformation confirms GLS results; $\rho = 0.033$
6	Refined Model	Temperature and Unemployment are robust predictors after removing Fuel_Price

This analysis has shown the entire process of identifying and adjusting the autocorrelation issue within a time series regression model. The Durbin Watson test has shown a statistically significant but minor first-order autocorrelation within the OLS residuals of the Walmart model. The Fuel_Price variable, which could be seen as a potentially significant predictor through naive OLS, is shown to be insignificant when the proper model and adjustment are used.

The major point is that autocorrelation, even when it is weak, should not be neglected, as it can lead to inference bias. The GLS Cochrane Orcutt method is successful in overcoming the AR(1) problem of autocorrelation, and reliable results are obtained. CPI and Unemployment are established as the strongest predictors of retail sales, which is logical from a macroeconomic perspective, as both are known to act as depressors of consumer spending.

References

1. IGNOU. Autocorrelation (Unit 7). eGyanKosh Study Material.
<https://egyankosh.ac.in/bitstream/123456789/78535/1/Unit-12.pdf>
2. IGNOU. Regression Models with Indicator Variables (Unit 9). eGyanKosh Study Material.
<https://egyankosh.ac.in/bitstream/123456789/88184/1/Unit-9.pdf>

Code File

https://drive.google.com/drive/folders/1X6RRvuDHqmwCVQLvfeREuMURFzkbN_u?usp=sharing